

Item Features Affect How Well People Can Predict Self-Reported Personality Traces of Strangers



Richard Rau^{1,2} , Sebastian Kueppers³ , Mitja D. Back^{2,4} and Michael P. Grosz¹

Abstract

Previous work suggests that self–other agreement in personality judgment is lower for evaluative than for neutral items. We tested whether this pattern generalizes to judgments collected with *Who Knows*, a mobile app in which users watch brief video clips of strangers and try to predict how targets answered highly concrete personality statements. The dataset comprised >340,000 predictions from >4,000 judges across 296 items and 75 targets. Contrary to prior findings, item evaluativeness did not reduce prediction accuracy with respect to targets' self-reports. Instead, evaluativeness was positively associated with prediction accuracy for Likert-type items while no effect was observed for binary items. Other coded item features (e.g., base rate and psychological modality) showed no consistent associations with prediction accuracy across formats. We propose two boundary conditions that may help reconcile these results with earlier work: (a) the present item pool is unusually concrete, which alters the distribution and correlates of desirability/evaluateness, and (b) judges were instructed to predict targets' self-reports rather than provide their own impressions. Under this instruction, judges may partly rely on between-target differences in self-protective responding when forming predictions. These findings caution against transferring item-feature effects from adjective-based impression paradigms to concrete items and self-report prediction tasks.

Plain Language Summary

Personality research often compares how people describe themselves with how others describe them. Earlier studies suggest that this self–other agreement tends to be lower for evaluative statements (e.g., “fair”) than for more neutral statements (e.g., “talkative”). We tested whether this pattern holds for highly concrete statements (for instance “afraid of spiders”) which people evaluated about strangers using the mobile app *Who Knows*. Within this app, users try to predict how people viewed in short self-introduction videos answered concrete statements about themselves. The dataset included over 340,000 predictions from more than 4,000 users, across 296 statements and 75 target persons. Contrary to earlier findings, evaluative statements did not reduce how well users could predict targets' self-reports. For rating-scale statements, evaluativeness was linked to slightly higher prediction accuracy; for yes/no statements, evaluativeness was unrelated to accuracy. Other statement features (such as whether a statement concerned thoughts, feelings, or actions, or whether it described something rare vs. common) showed no consistent links to prediction accuracy. We suggest two reasons why our results differ from prior self–other agreement studies. First, our statement set was unusually concrete, which changes the distribution and meaning of evaluativeness. Second, users were explicitly asked to predict self-reports rather than give their own impressions. Under these instructions, users may successfully infer how much targets tend to self-present in favorable ways. Overall, what is known about the effects of typical questionnaire statements from classic impression research may not transfer directly to concrete statements and self-report prediction tasks.

Keywords

self-report prediction accuracy, self–other agreement, personality judgment, item features, evaluativeness

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¹Institute for Mind, Brain, and Behavior, HMU Health and Medical University Potsdam, Potsdam, Germany

²Department of Psychology, University of Münster, Münster, Germany

³Department of Psychology, University of Hamburg, Hamburg, Germany

⁴JICE, Joint Institute for Individualization in a Changing Environment, University of Münster and Bielefeld University, Münster and Bielefeld, Germany

Corresponding Author:

Richard Rau, Institute for Mind, Brain, and Behavior, HMU Health and Medical University Potsdam, Schiffbauergasse 14, Potsdam 14467, Germany.

Email: richard.rau@hmu-potsdam.de

Imagine you have just talked to a new colleague for a couple of minutes or you have been briefly introduced to a friend's new partner at a party. Even though these interactions might not have felt particularly informative, you probably spontaneously formed a first impression of the person's personality such as how talkative or fair they are. Research suggests that, for relatively neutral personality aspects such as *talkative*, such inferences align well with the target person's self-report whereas for relatively evaluative aspects—those that describe a person in a generally favorable or unfavorable light to others—such as *fair*, they align much less (e.g., John & Robins, 1993). The predominant explanation for this in the literature has been that evaluative items invite self-protective biases in self-reports, reducing agreement with others' judgments (Vazire, 2010). However, more evaluative items also tend to be more abstract, thus allowing for more interpretative wiggle room in self-reports (Semin & Fiedler, 1988). As such, it is unclear whether evaluative item content would still undermine self–other agreement in a scenario where items are evaluative but also concrete rather than abstract. More generally, it remains unclear to what extent item-feature effects observed in the self–other agreement literature generalize across judgment tasks and instruction sets. Finally, beyond evaluativeness and abstractness, items differ in many other regards, for example, whether they describe rare versus common personality aspects or whether they describe feelings versus behavior. Previous research has not sufficiently examined these features' potential influences on self–other agreement.

The present study therefore pursues several goals. First, we examine whether the commonly reported negative association between evaluativeness and self–other agreement is also observed in a setting where (a) judgments are made regarding highly concrete items and (b) judges are explicitly instructed to infer how targets would answer about themselves. Second, we examine how a broader set of item features beyond evaluativeness and abstractness (e.g., base rate, psychological modality) covary and how they are jointly and uniquely associated with prediction accuracy. To this end, we analyze over 340,000 video-based first impression judgments collected via the quiz app Who Knows (Rau et al., 2025). In this task, judges deliberately try to predict targets' self-reports. The predictions pertain to a large pool of highly concrete items that are heterogeneous in evaluativeness and many additional features. We quantify success by estimating, for each item, the association between judges' predictions and targets' self-reports. In prior work, this association has typically been labeled *self–other agreement* but here, to make the criterion and task demands explicit, we refer to it as *self-report prediction accuracy* and use *prediction accuracy* as shorthand.

Previous Evidence Suggesting that Evaluativeness Undermines Self–Other Agreement

To our knowledge, four studies have directly examined whether a personality item's evaluativeness moderates the

association between targets' self-reports and others' judgments on that item. In these studies, the outcome is simply the item-level correlation between others' judgments and targets' self-reports. All four studies suggest that evaluativeness reduces this association. Specifically, John and Robins (1993) first reported a large negative association of $r = -.53$ both in a set of $n = 76$ adjectives (Study 1) and in a second set of $n = 80$ adjectives (Study 2). Second, a meta-analysis by Kenny and West (2010) spanning $n = 118$ items from 24 primary studies found a negative zero-order correlation of $r = -.35$ between evaluativeness and self–other agreement.¹ Third, De Vries et al. (2016) found modest but significant negative relations between evaluativeness and self–other agreement in two studies ($r = -.12$ and $r = -.21$ in sets of $n = 240$ and 200 items, respectively). Finally, a recent study by Wessels et al. (2025) found clear negative quadratic effects of social desirability on self–other agreement in two studies ($n = 107$ items and $n = 111$ items). Since evaluativeness values are a nonlinear transformation of social desirability values (i.e., medium-desirability corresponds with low evaluativeness, whereas both high and low desirability correspond with high evaluativeness), this finding conceptually aligns with a negative linear effect of evaluativeness, although its effect size cannot be expressed on the correlation metric. Taken together, although effect sizes vary, the literature provides consistent evidence that the association between targets' self-reports and others' judgments is weaker for evaluative than for neutral items. A noteworthy strength of this joint evidence is that it comes not only from participants who were well-acquainted (De Vries et al., 2016; John & Robins, 1993; Wessels et al., 2025) but also from casual acquaintances or even strangers (Kenny & West, 2010).

How can this effect be theoretically explained? A well-suited and formalized framework for this is offered by the *substance-and-evaluation model* (Leising et al., 2015, 2025), a generic version of which is illustrated in Figure 1 (Panel A). In this model, both self-judgments and other-judgments reflect (a) an underlying attribute component (S) that the item is intended to capture and (b) evaluative/attitudinal components (A) that can shape responses over and above this attribute component. The strength of these influences (the s and a paths) is assumed to vary across items (Leising et al., 2025). Most importantly, more evaluative items are theorized to show stronger a paths because they afford greater expression of unique attitudes toward the target. Because the self's and other's attitude components are typically only weakly aligned, stronger a paths reduce the correlation between self- and other-based responses (the x path). In other words, evaluative items can introduce variance that is not shared across self and other, thereby lowering self–other agreement.

A related, yet more specific explanation of how and why evaluativeness might undermine self–other agreement is offered by the self–other knowledge asymmetry (SOKA) model (Vazire, 2010). The SOKA model emphasizes that evaluative items may show low self–other agreement primarily because targets' self-reports are

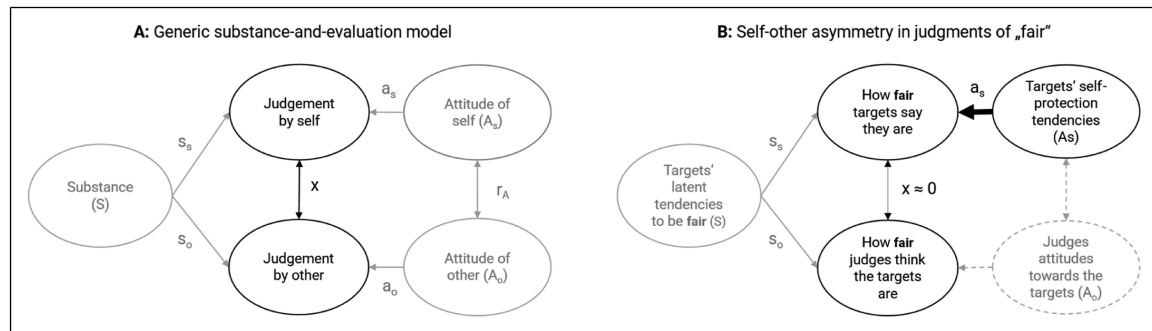


Figure 1. Existing explanations for the presence or absence of self–other agreement (x)

Note. Panel A shows a generic substance-and-evaluation model where self–other agreement (x) is jointly shaped by a substantive component (S) and evaluative/attitudinal components (A). Further, S may influence the A variables but these paths are omitted for simplicity. Panel B illustrates self–other asymmetry for an abstract, highly evaluative item (“fair”) where evaluative self-protection (A_s) strongly biases self-reports and where the other’s attitudes (A_o) are assumed to be invariant (cf. dashed elements), together undermining self–other agreement ($x \approx 0$)

strongly shaped by differential self-protective biases, whereas others’ attitudes towards the target are assumed to have a largely constant influence across target-judge dyads and to therefore be mostly irrelevant in many judgment contexts.² Panel B of Figure 1 illustrates this logic for judgments of *fair*, a highly evaluative adjective (Hampson et al., 1987), which exhibited essentially no self–other agreement in John and Robins’ seminal study ($r = .01$, John & Robins, 1993). According to the SOKA logic, the key source of disagreement regarding “fair” is a strong a_s path, at least to the degree that disagreement stems from high item evaluativeness rather than low item observability. In this panel, the A_o -related elements are shown as dotted because the specific SOKA-based explanation highlighted here treats them as comparatively invariant across targets and therefore not central to the focal reduction in self–other agreement.

Why Evaluativeness Might Not Always Undermine Self–Other Agreement

There are two important caveats to consider before accepting the claim that evaluativeness undermines self–other agreement as a generalized principle. The first caveat is that most empirical studies reporting this effect relied on single-descriptor adjective items (e.g., “fair”) and, relative to phrased items, adjectives are systematically more abstract (Semin & Fiedler, 1988), leaving more room for interpretive latitude and scale-use variability. For instance, the association between self and other might be undermined not just from self-reports being biased due to self-protection (i.e., by a strong a_s path) but also from ambiguity about what “fair” means, which would weaken the extent to which self-reports track the underlying attribute component (i.e., a weaker s_s path in Figure 1, Panel B). Figure 2 illustrates how self–other agreement might look different for evaluative but relatively concrete phrased items (low abstractness ratings in our item-feature data), using two examples from the present item pool: a scale item (“How likely is it that X secretly reads someone else’s diary?”; evaluativeness = 4.18; abstractness = 1.06) and a binary item (“Does X sometimes park briefly in a

disabled parking space?”; evaluativeness = 3.76; abstractness = 0.58). For such concrete items—and especially for dichotomous response formats—there may be less interpretive and scale-use latitude in self-reports, potentially strengthening s_s and attenuating the usual negative association between evaluativeness and agreement. Notably, even highly concrete descriptions can retain some interpretive flexibility depending on context and connotation. Our rationale targets relative differences in the degree of such flexibility across items rather than assuming any item is interpretation-free.

Self-Other Agreement Versus Self-Report Prediction Accuracy

The second caveat concerns task instructions. Most existing studies on self–other agreement asked judges for their personal impressions of the target. In contrast, judges can also be instructed to predict how targets would describe themselves, treating a response as accurate if it aligns with the target’s self-report. This instruction turns the task into a self-report prediction and, in principle, can change the processes that generate an association between judgments by the self and the other.

To our knowledge, only one study exists to date that has directly compared a standard impression instruction with a self-report prediction instruction. In this research (Solomon & Vazire, 2016), the responses to both types of instructions were very highly correlated, their means and standard deviations were essentially identical, and their correlation with targets’ self-reports was extremely similar. Although these findings do not directly speak to potential effects of item evaluativeness, they suggest that the type of framing in a personality judgment task might make less of a difference than one might intuitively think. Nevertheless, the self-report prediction instruction has a theoretically important implication, as illustrated in Panel B of Figure 3. Specifically, because the judgment from the other-perspective is deliberately aimed at aligning with the self-judgment, there is now possibly a direct effect of the target’s self-protection tendency to the other-judgment (see the path labeled y). In other words, judges may anticipate the extent to which targets’

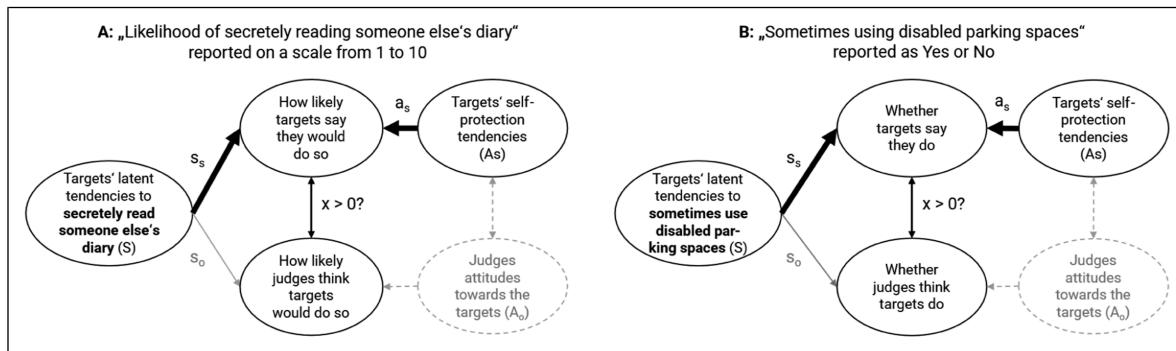


Figure 2. Illustration of Caveat 1: Concreteness increasing the substance saturation of self-reports

Note. For highly concrete items (scale vs. binary response formats), self-reports may track the substantive component (S) more strongly (stronger s_s), potentially attenuating the typical negative association between evaluativeness and self–other agreement (x). Line thickness reflects hypothesized effect size; dashed elements indicate paths/components omitted for simplicity

self-reports are shaped by evaluative self-protection (i.e., A_s). We are not aware of empirical research on whether judges are capable of this, but in principle, this seems possible. Specifically, to the degree that there exist valid cues indicating between-target differences in self-protection tendencies which are correctly utilized by judges, the y path would be positive and explain why higher item evaluativeness might even *promote* prediction accuracy when a self-report prediction instruction is used. The video-based format of Who Knows may make this route particularly plausible because judges could draw not only on abstract beliefs about the target, but also on visible cues to how the target presents themselves, including appearance, clothing, nonverbal behavior, and general self-presentation style.

Taken together, although the existing literature suggests that evaluativeness undermines self–other agreement, it is not clear whether the same pattern holds in a setup where (a) items are highly concrete and response formats reduce interpretive and scale-use latitude in self-reports and (b) judges are explicitly instructed to predict targets’ self-reports, potentially taking targets’ self-protection into account. The present research tests these boundary conditions using a unique dataset in which a very large number of judges evaluated a heterogeneous pool of unacquainted targets on many different items spanning a wide range of personality-relevant content. In addition, we examine how item features beyond evaluativeness relate to prediction accuracy.

Item Features Beyond Evaluativeness

So far, we have discussed three features in which items might vary: Abstractness, response format, and evaluativeness. However, there are many additional ways in which items might differ from one another. Specifically, we consider the items’ *observability*, *psychological modality*, *timeframe*, *baserate*, *baserate extremity*, *interpersonal importance*, and *value-relatedness*.

Observability and Psychological Modality

Observability has been one of the most commonly examined features and is typically positively related to self–other

agreement (Kenny, 1994). Operationally, it is often defined in terms of how “hidden” versus “observable” item content is from an outside perspective (see Appendix A). More formally, observability can be understood within the substance-and-evaluation model (Leising et al., 2025) in terms of the path s_o , that is, by how strongly substantive target differences manifest in the judgments of others. As such, observability is technically not an item feature but rather a feature of the combination of item and judgment context. For instance, the item “secretly reads others’ diaries” is not per se observable or unobservable but is observable in some judgment contexts (if the judge has an opportunity to witness the target reading or resisting reading someone’s diary) but unobservable in many other contexts (if no such opportunity is offered in a judgment context). Reflecting this confusion, operationalizations of observability have differed substantially in previous research, often blending item properties with judgment-context properties (see Appendix A) despite its framing as a mere item (or trait) feature. To keep the feature definitions item-centered and comparable across contexts, we here complement observability with an item’s *psychological modality*—whether the item refers primarily to affect (A), behavior (B), cognition (C), or desires (D; Wilt & Revelle, 2015). Because overt behavior is, by definition, more directly observable than affect, cognition, or desires, modality may capture variance in prediction accuracy that earlier work has attributed to observability.

Time Frame

Items differ in terms of the time frame they refer to. Specifically, they may refer to behavior that has happened in the past (e.g., “has X ever made out at the movies?”) which includes references to residues of past behavior (e.g., “does X have a tattoo?”), or they may refer to present or habitual behavior (e.g., “does X write to-do-lists?”) or to future or hypothetical behavior (e.g., “would X accept an invitation to go bungee jumping?”). The concept of an item’s time frame has been used in personality research in a broad sense (Werner & Pervin, 1986) but has to our knowledge not been considered in research on self–other agreement.

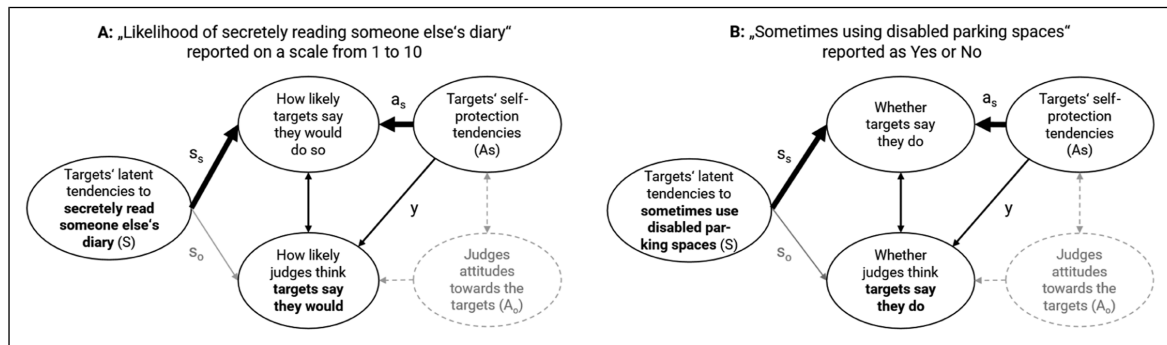


Figure 3. Illustration of Caveat 2: Self-report prediction instructions add a direct pathway (y)

Note. Under a self-report prediction instruction, judges' inferences may reflect not only the substantive component (S) but also anticipated evaluative self-protection in targets' self-reports, adding a direct pathway (y) from targets' self-protection tendencies to judges' inferences and potentially changing how evaluativeness relates to the association between judgments by the self and the other (x). Line thickness reflects hypothesized effect size; dashed elements indicate paths/components omitted for simplicity

Baserate and Baserate Extremity

Some items describe aspects that are self-reported very frequently in the general population and others describe aspects that are rarely ever reported. These differences in items' baserates have been examined in the context of self-other agreement for the first time by Wessels et al. (2025) but have not emerged as a consistent antecedent of self-other agreement. However, we agree in principle with Wessels et al.'s (2025) reasoning that extreme baserates (i.e., either very high or very low endorsement rates across targets) would restrict the variance in judgments and could thereby possibly undermine agreement. As such, we follow their example and consider items' raw baserates as well as the extremity of those baserates.

Interpersonal Importance

It has been suggested that some personality traits of potential interaction partners are more important to know about than others (Wood, 2015). Wessels et al. (2025) included importance in their studies and found it to be negatively related to prediction accuracy, meaning that items that were consensually seen as more important produced less self-other agreement. Here, we test whether this somewhat surprising finding holds in our novel item pool.

Value-Relatedness

Finally, we consider items' relatedness to *values* in our analysis. Values denote "moral, social, or aesthetic principle(s) accepted by an individual or society as a guide to what is good, desirable, or important" (VandenBos, 2015). Although value-relatedness is likely correlated with interpersonal importance and evaluativeness, we included it as a separate feature for comparability with previous research on the question of assumed similarity (when do judges assume targets to be similar to themselves?) were value-relatedness has emerged as a particularly relevant item feature (Thielmann et al., 2023).

The Present Research

The present research examines whether item evaluativeness undermines prediction accuracy in a judgment context where (a) targets' self-reports are likely less biased than in previous research because of reduced ambiguity in item meaning and scale-use and where (b) judges are explicitly prompted to take a meta-perspective and predict how the target would self-report. In addition to this critical test, a second goal is to explore and document how item features other than evaluativeness contribute to prediction accuracy. Documenting the distributions of and covariances among these features adds to the generally quite scarce literature on item features. To our knowledge, the few existing studies that have provided such documentations have either focused on pre-defined item sets from circumscribed personality inventories (e.g., De Vries et al., 2016; John & Robins, 1993) or on comprehensive lists of adjectives (Britz et al., 2019, 2023; Leising et al., 2014). By providing a resource covering a very large and heterogeneous list of phrased items, we hope to contribute a valuable resource for future research.

To achieve these goals, we analyze inferences about actual people that were collected in a large-scale, gamified online assessment via the Who Knows app (Rau et al., 2025). These inferences were made by 4,025 different users and pertain to 75 different targets. Most crucially, they concern 296 highly concrete items which jointly cover a plethora of personality manifestations and personality traces.

Methods

Who Knows is a free web-application with which users can test their ability to predict how strangers would self-report about a variety of concrete personality-related aspects. Details about the project are documented in Rau et al. (2025) and an exemplary game round within the app is illustrated in Figure 4. In essence, users get to view a short video of a target person introducing themselves for about 30 seconds and then they make judgments about that

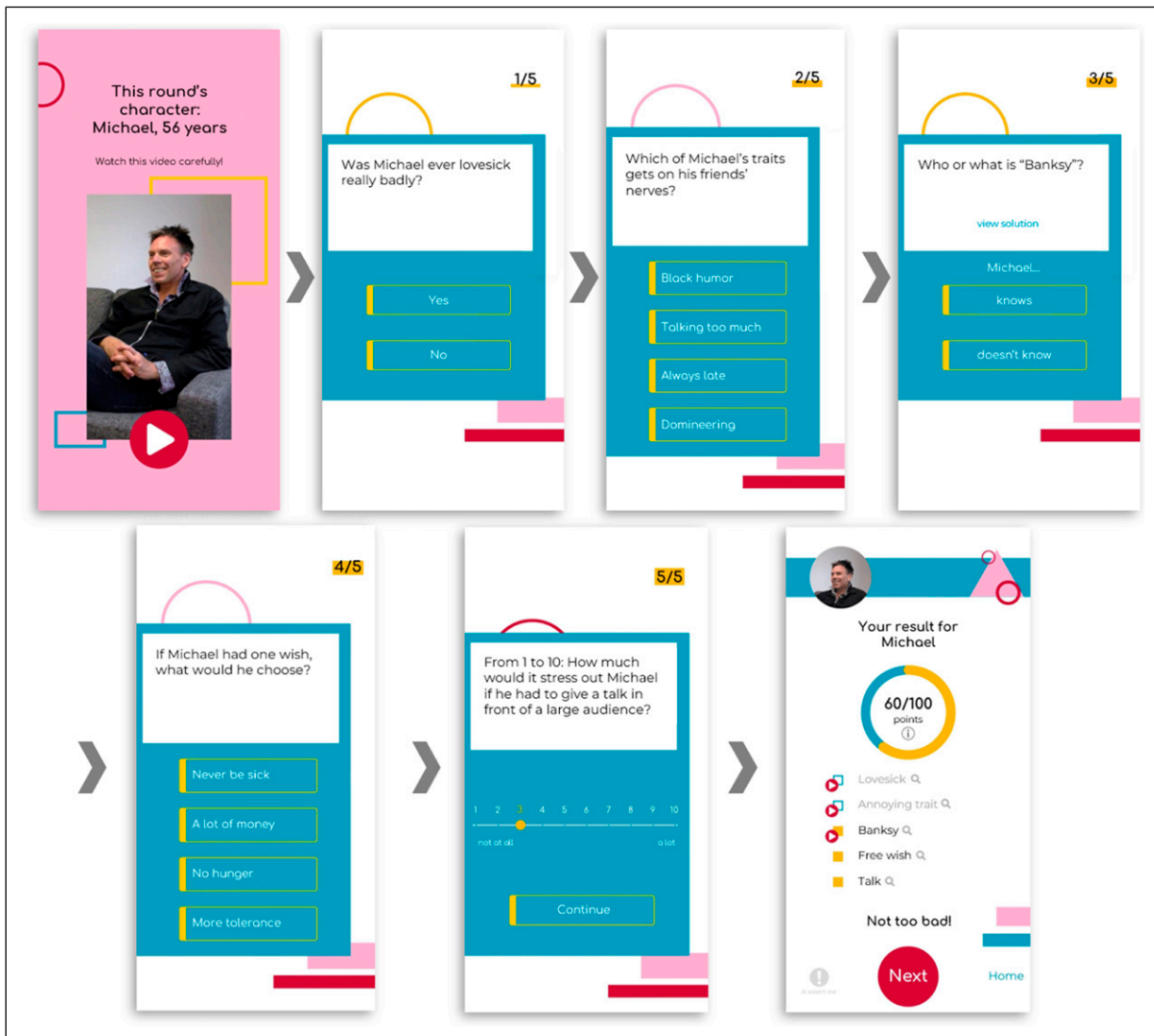


Figure 4. Exemplary game screens of the Who Knows app

Note. Hitting the play button in the top left screen starts the target's introduction video. The following five screens record the user's judgments regarding random items that have been answered by the target. The bottom right screen displays the user's results for that round. All texts are free translations from the German original. Items in the second, third, and fourth game screen were not included in the main analysis

target on five distinct items. After this, they get feedback as to which of their judgments matched the target's self-report. The next round will show a novel target and (very likely) present five novel items. This is because, whenever a user starts a game round in the Who Knows app, one of the targets which are available in the database of the app gets selected (pseudo-)randomly and for the selected target, five of the available items get selected (pseudo-)randomly. Users can play whenever they like and essentially as much as they like (up to ~800 rounds). To encourage high user engagement, the app's interface follows modern design principles and is highly gamified (e.g., users collect points, earn medals, and are granted experience levels).

The data collection is ongoing but for the present study, we analyze the data that have been collected as of August 2024. We excluded responses from users who played fewer than 10 rounds to facilitate model convergence (see Analyses section below). After this exclusion, the data contained

344,929 judgments collected from 4,025 users who judged 75 targets on 296 items. This study was not preregistered.

Items

The item pool was generated by four research assistants who were instructed to brainstorm for a pool as large and heterogeneous as possible. The primary criterion was to cover anything that could be interesting or fun to learn about a stranger. Importantly, the item pool was not designed as a psychometric instrument and was not intended to represent (or recover) the Big Five/FFM structure. Instead, items were generated to maximize breadth and entertainment value, and we treat them as a heterogeneous set characterized by item features rather than by any assumed trait taxonomy. As such, the research assistants not only sought inspiration from comprehensive personality inventories but also from TV shows and board games on interpersonal perception.

Table 1. Example items for different feature levels

Feature: Level	Example item (feature value)
Abstractness: Minimal	Does X check the change when shopping groceries? (0.26)
Abstractness: Maximal	How imaginative is X? (3.37)
Desirability: Minimal	How likely is it that X would start a bar fight? (−4.17)
Desirability: Maximal	How much does X like children? (4.45)
Evaluativeness: Minimal	Does X own more than one credit card? (0.83)
Evaluativeness: Maximal	How punctual is X? (4.71)
Observability: Minimal	How often does X think about sex per day? (0.89)
Observability: Maximal	Does X have tattoos? (4.41)
Psychological modality: Affect	Does X get scared when he/she watches crime thrillers?
Psychological modality: Behavior	Does X chew on pencils?
Psychological modality: Cognition	Can X get excited about classical paintings in an art museum?
Psychological modality: Desire	Would X like to be on the cover of a famous magazine?
Timeframe: Past	Has X ever secretly read the text messages of a partner?
Timeframe: Present	Does X play an instrument?
Timeframe: Future	Would X feel comfortable loudly shouting “Hello” across a large public square?
Baserate: Minimal	Does X always have a few stamps in his wallet? (0.74)
Baserate: Maximal	Does X wear sweatpants at home? (4.41)
Baserate extremity: Minimal	Does X use placemats when eating? (1.3)
Baserate extremity: Maximal	Does X wear sweatpants at home? (3.81)
Interpersonal importance: Minimal	Does X like the photo on his/her ID card? (0.37)
Interpersonal importance: Maximal	How well can X keep a secret? (3.89)
Value-relatedness: Minimal	Has X ever locked herself/himself out of her/his home? (1.03)
Value-relatedness: Maximal	How important are the feelings of others to X?

They were also encouraged to generate concrete rather than abstract items and to pick for each item a response format (binary yes-or-no question vs. open-ended question with 4 response options for judges vs. scale-rating from 1 to 10) which they found to fit the item best. Items from this pool were checked for spelling mistakes, ambiguities, redundancies, etc. and corrected or removed accordingly. All remaining items were kept (i.e., none were excluded based on psychometric properties). This yielded 811 items, 350 of which were in the formats considered in the present study (137 rating-scale items and 213 binary forced-choice items).³ Of these, we included only those items that had been judged with nonzero variance by the users and had been answered by at least five different targets with nonzero variance to avoid conflating item difficulty with target readability. This yielded a final sample of $n = 296$ items (123 rating-scale items and 173 binary forced-choice items). Example items are shown in Table 1 and a complete list of the items and their features can be found on the OSF (<https://osf.io/rjs2u/>).

Targets

The 75 targets occurring in the Who Knows app were recruited via flyers, wallpapers, mailing lists, and local newspaper announcements. They were aged between 18 and 73 ($M = 39$, $SD = 17$) and 55% of them identified as female, 44% as male, and 1% as diverse. They were interviewed in a video laboratory and provided informed consent for using their first name, age (at the

point of the interview) and video material in a public web-application. Some targets were interviewed in the summer of 2021 ($n = 53$) and others were interviewed in the spring of 2023 ($n = 22$). In their interview, targets were asked to answer a random selection of 100 items drawn from the item pool and they could skip as many items as they liked. To encourage honesty, they were told that their participation offered them the rare opportunity to receive genuine insights into the impressions they make on strangers and that this feedback would be most valuable if they behaved as authentically as possible during their interview. Specifically, we offered each target personalized, item-level feedback showing frequency plots of all judgments made about them in Who Knows app.

Judges

The users of the Who Knows app are recruited mostly via social media posts, flyers, occurrences in public media such as podcasts and magazines, and passively via word of mouth. During their account setup, users are informed that their participation is voluntary and that their anonymous data will be used for scientific purposes. Upon signing up, users can play up to 10 rounds. They then have to provide sociodemographic information before being able to continue playing after the 10th round. The 4,025 users considered in the present analysis were aged between 18 and 88 years ($M = 39$, $SD = 14$) and 73% identified as female, 25% as male, 1% as diverse, and 1% preferred for their sociodemographic information not be used.

Item Features

The level of abstractness, social desirability, and value-relatedness of all items were rated in an online survey in return for course credit by 29 undergraduate students (24 female, 4 male, 1 diverse; ages between 18 and 26 years), all of whom were native German speakers. Observability, baserate, and importance were similarly rated by another group of 27 individuals sampled from the same population (24 female, 1 male, 2 diverse; ages between 18 and 37; 9 individuals overlapping with the first group of raters). One individual in this group was a non-native but fluent German speaker and all remaining individuals were native German speakers. All raters in both groups were majoring in Psychology. Items were presented one-by-one to minimize contrast effects and were presented as statements (e.g., “X likes to dress up,” “X likes children a lot”) instead of questions (e.g., “Does X like to dress up?”, “From 1 to 10: How much does X like children?”) to avoid influences of the response format. Items were presented in four sets, each containing a quarter of the items in a fixed random order. Rating a set of items took approximately 20 minutes and the raters were forced to take a break of at least 1 minute before they could continue with the next set. Set order was reversed for half of the participants to counterbalance order effects. Details on the rating instructions can be found under <https://osf.io/rjs2u/> (S1). Inter-rater agreement was high for all features except abstractness (ICCs[3, k] ranging from .87 to .92). For abstractness, inter-rater agreement was initially somewhat borderline, ICC(3, 29) = .70, but was satisfactory after we excluded ten raters whose ratings suggested that they might have misunderstood the rating instruction⁴, ICC(3, 19) = .79.

Evaluativeness values were computed as the absolute values of the mean social desirability ratings, which equals the absolute difference from the midpoint of the social desirability scale. Values for baserate extremity were computed as a nonlinear transformation of the mean baserate ratings (e.g., transforming baserate values of both 0 and 5 into baserate extremity values of 5). The categorical item features of psychological modality and timeframe were classified by a trained research assistant where any instances of ambiguity were discussed with the first author and resolved consensually.

Analyses

Our analyses involved three steps. First, we aggregated the ratings of the item features across raters and determined the features' reliability. Second, we indexed the level of prediction accuracy with which each item was judged by an average Who Knows user. Third, we used the item features to predict the observed level of prediction agreement. We next elaborate on the second and third step in more detail.

To index item-level prediction accuracy, we estimated two mixed-effects regression models, one for rating-scale items (linear) and one for binary forced-choice items (logistic). In both models, each individual judgment (a given

judge's response about a given target on a given item) was regressed on the target's self-report for that item (see Biesanz, 2010; Kueppers et al., 2024, for a similar modeling logic). Prediction accuracy is reflected in the extent to which judges' responses track between-target differences in targets' self-reports; accordingly, positive slopes indicate higher prediction accuracy and slopes near zero indicate no systematic alignment with targets' self-reports.

In each model, we allowed the effect of targets' self-reports on judges' responses to vary across items by specifying an item-level random slope for the self-report predictor.⁵ Item-specific slopes were obtained as the sum of the fixed-effect slope and the item-specific random deviation (i.e., fixed + random slope for each item). These item-specific slopes were then transformed to a common correlation-like metric (r) to enable comparability across response formats.

For rating-scale items, we expressed item-specific linear slopes on an r -like metric by first converting each item-specific slope to a t -value using its conditional standard deviation from the fitted random-effects structure and then applying the algebraic t -to- r transformation $r = \frac{t}{\sqrt{t^2 + df}}$ (Rosenthal & Rosnow, 2008). Specifically, $t = b/SE(b)$, where $SE(b)$ was obtained from the conditional (posterior) variance of the item-specific random slope returned by the mixed model, and df was approximated as the number of usable observations for that item minus one. We treat the resulting values as a correlation-like rescaling that places item effects on a familiar metric.

For binary items, we converted item-specific logistic slopes (b) to a latent-normal effect size using $d = b \frac{\sqrt{3}}{\pi}$ (Chinn, 2000), and then expressed this effect on a correlation-like scale while accounting for imbalance in the binary self-report predictor. Specifically, for each binary item we computed the self-report base rate $p_x = P(X = 1)$ across targets and used $A = 1/(p_x(1 - p_x))$ to obtain $r = \frac{d}{\sqrt{d^2 + A}}$.

Finally, for both response formats, we quantified uncertainty of the transformed r values via the delta method by propagating the conditional variance of the item-specific random slopes through the respective transformation functions. These standard errors were used only for precision-weighted robustness analyses reported in the supplement; all primary analyses treat the item-level r values as the outcome without additional weighting.

Importantly, the resulting index of prediction accuracy has two desirable properties: First, it reflects the level of prediction accuracy that is expected when a target is judged by a *single* (typical) judge. Alternatively, previous research has sometimes also computed the association between targets' self-reports and the average judgment of *multiple* judges. However, due to the aggregation, this index increases with an increasing number of judges (Back & Nestler, 2016) and is therefore ill-suited for comparing the results of studies with varying numbers of judges. Second, our index does not measure *absolute* but *relative* prediction accuracy. For instance, if all targets were to report very low values on the item “how likely are you to secretly read someone else's diary” (e.g., only 1 s,

2 s, and 3 s on a scale from 1 = *not likely at all* to 10 = *very likely*), and judges were to correctly use only 1 s, 2 s, and 3 s in their judgments of the same item, this alone would minimize the absolute, numerical distance between the self- and the other-judgment, but would not automatically yield a high value on our self-report prediction accuracy index. Instead, judges would need to detect who of the targets expresses a particularly low likelihood for reading someone else's diary (i.e., reporting 1 s) relative to other targets (i.e., reporting 2 s or 3 s). With these properties, our index mirrors what has sometimes been coined *distinctive accuracy* (as opposed to *normative accuracy*) in the literature on judgmental accuracy (Biesanz, 2010) and compares well to the way self-other agreement has been indexed in previous research on item-level moderators of agreement (De Vries et al., 2016; John & Robins, 1993; Kenny & West, 2010; Wessels et al., 2025).

Finally, in Step 3, we used the item features from Step 1 to predict the items' prediction accuracy levels from Step 2 using classic linear regression. We first did this by considering each item feature as the sole predictor in a separate model. The regression coefficients of these single-predictor models should be causally informative for two reasons. First, reverse-causation (prediction accuracy influencing the item feature) is logically impossible. Second, we believe that none of the studied item features (e.g., abstractness) causally precedes any other item feature (e.g., evaluativeness). Thus, none of the item features can be a potential confounder of the effect of another item feature on accuracy, as a confounding variable needs to cause both the treatment and outcome variables (e.g., Wysocki et al., 2022). Nevertheless, we anticipated possible associations among the diverse item features, and also ran a multiple regression model including all features at once to learn about the unique, descriptive effects of individual features after accounting for their overlap with the remaining features.

As a robustness check, we also re-ran the above-described regression analyses while accounting for the uncertainty of the slope estimates that went into our prediction accuracy index during Step 2. On the one hand, this seemed indicated because the items differed in how often they had appeared in the Who Knows app and therefore varied regarding how precisely prediction accuracy had been indexed for them. On the other hand, the statistical literature has warned that precision-weighting may illegitimately shrink individual slope estimates a second time, after shrinkage has already occurred during the estimation of the random effects model (Raudenbush & Bryk, 2002; Robinson, 1991). Therefore, we report unweighted analyses in the main text and precision-weighted alternatives as a robustness check in the supplement. We achieved precision-weighting by computing standard errors of the prediction accuracy index via the delta method (Oehlert, 1992) using the derivative of the transformation with respect to the raw slope. Finally, to obtain a measure of precision rather than imprecision and to counter skewness that was produced by a small number of extremely precise estimates, the standard errors were

transformed to logarithmic inverse-variance weights by the formula $\log(\frac{1}{SE^2})$.

Finally, we also explored models which included all possible pairwise interaction terms between the features (e.g., interactions between abstractness*desirability, base-rate*timeframe, etc.). Data and R code to reproduce all results and can be found on the OSF (<https://osf.io/rjs2u/>).

Results

Descriptive Results for Core Item Features

Recall that it was an explicit goal for the item pool of the Who Knows app to include mostly concrete rather than abstract items. As can be seen in the top left panel of Figure 5, abstractness was indeed generally quite low and only 9 of the 296 items (<4%) had a value above the scale-midpoint.⁶ As further shown in Figure 5 (top right panel), the item pool tapped undesirable, neutral, and desirable content, producing a roughly unimodal distribution around the scale midpoint of neutral valence. As a consequence, evaluativeness values (which are absolute social desirability values) also varied across items but with a left-skewed distribution indicating more items near the point of neutrality than near maximal evaluativeness. Finally, the bottom right panel of Figure 5 shows the distribution of self-report prediction accuracy values which are on the correlation metric. All values were positive, suggesting that no item showed systematic misalignment between judgments by the self and the other. Further, prediction accuracy ranged from being negligibly small (i.e., near zero) to being appreciably but moderately sized (i.e., near .25). Descriptively, Likert-scale items varied more than did binary forced-choice items. The overall rather modest level of prediction accuracy suggests that the task faced by users of the Who Knows app, that is, judging highly concrete personality traces upon a brief self-introduction, is a quite difficult one. Nevertheless, prediction accuracy rates were clearly above zero and, more importantly, they differed from item to item.

Does Evaluativeness Undermine Prediction Accuracy in Who Knows?

Figure 6 shows how item evaluativeness and prediction accuracy were associated across items. Contrary to the pattern typically reported in the self-other agreement literature, evaluativeness was not negatively related to prediction accuracy in the present data. Instead, prediction accuracy tended to be higher for more evaluative items. This association was visible in the full item pool (left panel of Figure 6) but was driven by the subset of Likert-scale items, which showed a clear positive relation (middle panel). In contrast, no comparable relation was evident for binary forced-choice items (right panel). To illustrate, judges were better able to infer targets' self-reported likelihood of secretly reading someone else's diary (1–10; a highly evaluative item) than to infer targets' self-reported enjoyment of shopping (1–10; a comparatively

neutral item). Yet, judges' ability to tell whether targets reported sometimes using disabled parking spaces (Yes/No; a highly evaluative item) was not higher than their ability to tell whether targets reported liking to dress up (Yes/No; a comparatively neutral item). An interactive display of these results is available online (<https://purl.org/item-features/app>), allowing users to inspect the specific item texts of each data point.

Although the Who Knows task differs from prior work on item evaluativeness and self–other agreement (e.g., because it uses highly concrete items and instructs judges to predict targets' self-reports), the positive association we observed for Likert-scale items was unexpected. We anticipated that greater item concreteness would increase the substance saturation of self-reports and thereby attenuate the usual negative association between evaluativeness and agreement (prediction accuracy under a self-report prediction instruction). However, this rationale would not by itself predict *higher* agreement for more evaluative items. One possibility, consistent with the process implication illustrated in Figure 3 (γ path), is that under a self-report prediction instruction judges may partly anticipate between-target differences in self-protective responding, which could increase prediction accuracy specifically for evaluative content.

To probe this account, we conducted an exploratory analysis examining whether targets differed

systematically in the favorability of their self-reports and in the favorability of the self-reports that judges predicted for them, and whether these target-level tendencies were aligned. Here, we reserve “desirability” for an item-level characteristic (how socially desirable it is to endorse an item; signed), and we use “favorability” for the response-level product of desirability and a given response (or predicted response). If judges pick up on stable between-target differences in self-protection, then targets who provide more favorable self-reports on average should also elicit more favorable self-report predictions on average. Because such target-level response tendencies carry evaluative implications primarily for desirable or undesirable items, this alignment could help explain why evaluativeness related positively to prediction accuracy in the Likert-scale items.

To operationalize response-level favorability, we multiplied each self-report and each judgment by the item's desirability value. We then estimated crossed random-intercept models (no predictors) for self-report favorability and for judgment favorability and extracted target ICCs and target random intercepts. The results are summarized in Table 2. Targets showed systematic differences in the average favorability of their self-reports across items and, less strongly, in the average favorability of the judgments they elicited. Moreover, these target-level components were positively correlated, indicating

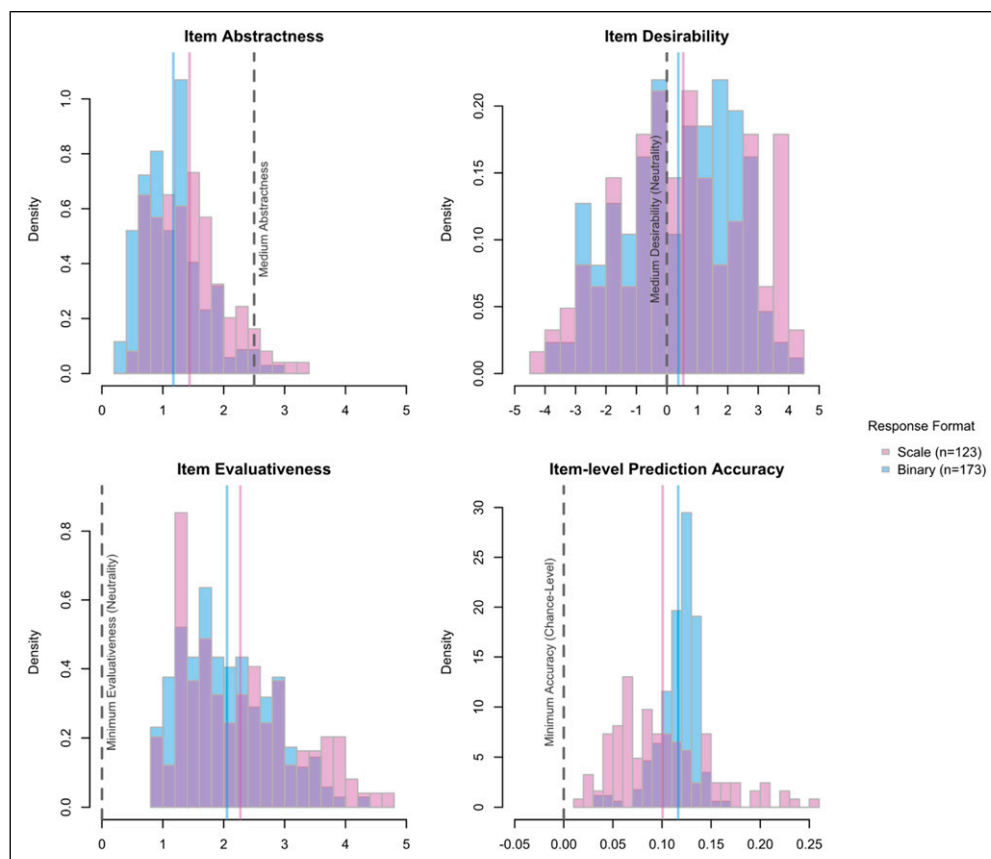


Figure 5. Distributions of abstractness, desirability, evaluativeness, and prediction accuracy
 Note. The distributions' means are shown as solid, vertical lines

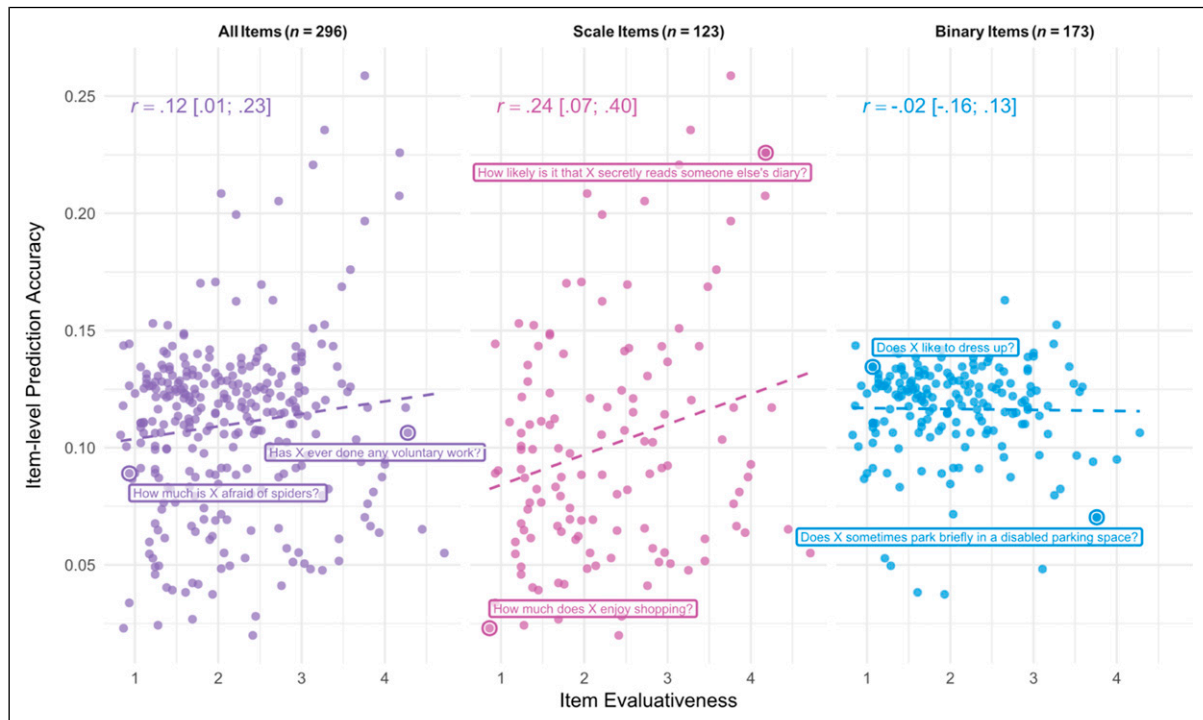


Figure 6. Effects of item evaluativeness on the level of self-report prediction accuracy by response format

Note. Prediction accuracy increases moderately with increasing item evaluativeness in the combined item pool (left panel). Splitting the item pool by response format shows that the effect is clearly present in Likert-type rating scale items (middle panel) but absent in binary forced-choice items (right panel). An interactive version of this plot can be viewed online (<https://purl.org/item-features/app>) where item texts are displayed dynamically as the user hovers over the respective data point

that targets with more favorable self-reports also tended to be targets for whom judges predicted more favorable self-reports. This pattern is compatible with the interpretation that judges used cues to targets' general self-protection tendencies when forming self-report predictions. Descriptively, this alignment was more pronounced for Likert-scale items than for binary forced-choice items, paralleling the fact that the positive association between evaluativeness and prediction accuracy was evident for Likert-scale items but not for binary items.

How Do Item Features Other Than Evaluativeness Impact Prediction Accuracy?

Table 3 shows descriptive statistics for all item features. The means for most of the continuous features other than abstractness were near the scale midpoint (i.e., between 2 and 3), the only exception being the feature of interpersonal importance being somewhat lower. All continuous features were distributed approximately normally. In terms of the discrete item features, a clear majority of items tapped the psychological modality of behavior and this was more pronounced in the Likert-scale items than in the binary forced-choice items. Across response formats, few items tapped desires. Further, most items pertained to the present rather than the past or the future. Among those items that did pertain to the future, most were Likert-scale items and among those that pertained to the past, most

were binary. These differences likely arose because the research assistants who generated items were instructed to pick response formats they subjectively found to fit item content best. Generally, the distributions of item features are unlikely to generalize to random samples of personality items because our items are less abstract than most personality items.

Table 4 shows the effects of all considered item features as single predictors in a regression with self-report prediction accuracy as the dependent variable, both using the complete item pool and using Likert-scale and binary forced-choice items in separation. Interestingly, no item features other than evaluativeness was predictive of prediction accuracy. This contrasts earlier work which consistently found positive effects of observability on prediction accuracy in more abstract item sets (De Vries et al., 2016; John & Robins, 1993; Kenny & West, 2010; Wessels et al., 2025) and occasional findings of positive effects of desirability and negative effects of interpersonal importance (Wessels et al., 2025).

Because some item features are logically intertwined, they might covary. For example, if an item is highly interpersonally important, it seems quite plausible that this item should also be related to values to a certain degree. As another example, if an item pertains to the future, it is necessarily hypothetical and therefore, its observability should be limited. Table 5 shows all associations among the item features and reveals several noteworthy results. First, there was indeed a cluster of consistently very high

Table 2. Exploratory analysis on favorability differences and their alignment

Exploratory research question	Result	
	Scale items	Binary items
(a) How much targets systematically differ in the favorability of their responses across items	ICC = .150	ICC = .078
(b) How much targets systematically differ in the favorability of the self-report predictions made about them across items	ICC = .040	ICC = .017
(c) How much (a) and (b) correlate	$r = .49$	$r = .31$

correlations among the features of evaluativeness, interpersonal importance, and value-relatedness ($.60 \leq r_{raw} \leq .67$; $.65 \leq r_{disattenuated} \leq .74$). Thus, highly evaluative items very often seem to capture content most people would find important to learn about as they get to know someone and they very often seem to relate to personal values.

Another noteworthy cluster of associations concerns the categorical variables of psychological modality and timeframe, both of which were associated with several other item features. The largest association of psychological modality was with abstractness and resulted mainly from cognition-related items being more abstract than behavior-related items. Another noticeable association was with observability. This association resulted mostly from items tapping desires being less observable than items tapping affect, behavior, and cognition (see S3 for more details).

The largest association of timeframe was with observability and resulted from items relating to the present being more observable than items relating to the past or future. Another noteworthy association of timeframe was with baserate. This association resulted from future-related items having lower baserates than those relating to the past or present (see S4 for more details). Finally, the two categorical features of psychological modality and timeframe were also substantially associated with each other. This resulted mainly from the facts that cognition-related items almost exclusively referred to the present and that desire-related items almost exclusively referred to the future (see S5 for more details).

Overall, the considered features show many substantial correlations which highlights the necessity to examine how they contribute to prediction accuracy not only in isolation but also above and beyond each other. At the same time, we deemed evaluativeness, interpersonal importance, and value-relatedness to be conceptually similar enough and empirically associated heavily enough, to warrant merging these variables into a common composite score rather than entering them separately in a multiple regression model. The results of the multiple regression analysis are shown in Table 6, both for the entire item pool and separately for Likert-scale and binary forced-choice items.

The multiple regression models revealed several interesting results. First, splitting the item pool by response format was once again instructive. Whereas no feature had a statistically significant effect on prediction

accuracy in the overall item sample, distinct effects became visible in the separated subsamples: In the subsample of Likert-scale items, the composite capturing the common core of evaluativeness, interpersonal importance, and value-relatedness was quite predictive of prediction accuracy, mirroring the single-predictor result for evaluativeness discussed above. Thus, judges reached greater prediction accuracy when they judged personality aspects that were more evaluative, important, and related to values, irrespective of whether these aspects pertained to particular psychological modalities, timeframes, had a particular baserate distribution, etc. In a sensitivity analysis in which the constituents of the composite were entered separately, only evaluativeness but not interpersonal importance and value-relatedness had a significantly positive effect on prediction accuracy. Further, in contrast to the zero-order results, observability emerged as a significant *negative* predictor in the multiple regression model. Observability showed a negative unique association in the multiple regression for scale items (Table 6), despite being nonsignificant in zero-order models (Table 4). We return to interpretation in the Discussion.

In the subsample of binary forced-choice items, the only feature with unique explanatory value was baserate extremity, which showed a significant, negative effect above and beyond the remaining item features. This suggests that for items for which a clear majority of targets can be expected to respond with “yes” rather than “no” (or vice versa), the judgments made by the Who Knows users tended to align less with the targets’ self-reports than for items with expected baselines closer to a 50–50 split. Since extreme baselines imply restricted variance and restricted variance implies a reduced ability to covary with an outside criterion, this effect is not surprising. Indeed, the baserate-variance relation is fully deterministic in case of a binary variable but not in case of a continuous variable. As such, the fact that we observed a negative effect of baserate extremity only in the subsample of binary items but not in the subsample of scale items supports this reasoning.

Additional Analyses

The robustness check with a precision-weighted regression model yielded largely identical results compared to the non-weighted main analyses. Most importantly, in the single-predictor models, evaluativeness was once again a

Table 3. Descriptive statistics of item features

Feature	All items (<i>n</i> = 296)	Scale items (<i>n</i> = 123)	Binary items (<i>n</i> = 173)
Abstractness	<i>M</i> (<i>SD</i>) = 1.28 (0.56)	<i>M</i> (<i>SD</i>) = 1.44 (0.59)	<i>M</i> (<i>SD</i>) = 1.17 (0.51)
Desirability	<i>M</i> (<i>SD</i>) = 0.44 (1.95)	<i>M</i> (<i>SD</i>) = 0.54 (2.08)	<i>M</i> (<i>SD</i>) = 0.38 (1.85)
Evaluativeness	<i>M</i> (<i>SD</i>) = 2.15 (0.84)	<i>M</i> (<i>SD</i>) = 2.27 (0.94)	<i>M</i> (<i>SD</i>) = 2.05 (0.75)
Observability	<i>M</i> (<i>SD</i>) = 2.56 (0.74)	<i>M</i> (<i>SD</i>) = 2.56 (0.71)	<i>M</i> (<i>SD</i>) = 2.56 (0.76)
Psychological modality	Affect: 14% Behavior: 56% Cognition: 24% Desire: 5% Not classified: 1%	Affect: 24% Behavior: 37% Cognition: 33% Desire: 7% Not classified: 0%	Affect: 7% Behavior: 69% Cognition: 18% Desire: 4% Not classified: 2%
Timeframe	Past: 19% Present: 61% Future: 20% Not classified: 0%	Past: 6% Present: 63% Future: 32% Not classified: 0%	Past: 29% Present: 60% Future: 11% Not classified: 0%
Baserate	<i>M</i> (<i>SD</i>) = 2.98 (0.65)	<i>M</i> (<i>SD</i>) = 3.00 (0.66)	<i>M</i> (<i>SD</i>) = 2.97 (0.64)
Baserate extremity	<i>M</i> (<i>SD</i>) = 2.21 (0.49)	<i>M</i> (<i>SD</i>) = 2.19 (0.46)	<i>M</i> (<i>SD</i>) = 2.23 (0.50)
Interpersonal importance	<i>M</i> (<i>SD</i>) = 1.75 (0.75)	<i>M</i> (<i>SD</i>) = 2.06 (0.76)	<i>M</i> (<i>SD</i>) = 1.54 (0.67)
Value-relatedness	<i>M</i> (<i>SD</i>) = 2.43 (0.62)	<i>M</i> (<i>SD</i>) = 2.64 (0.62)	<i>M</i> (<i>SD</i>) = 2.28 (0.57)

Table 4. Item features' influences on self-report prediction accuracy by response format

Predictor	All items (<i>n</i> = 296)		Scale items (<i>n</i> = 123)		Binary items (<i>n</i> = 173)	
	<i>b</i>	<i>p</i>	<i>b</i>	<i>p</i>	<i>b</i>	<i>p</i>
Abstractness	−.05	.354	.01	.888	−.04	.626
Desirability	.04	.544	.02	.868	.12	.131
Evaluativeness	.12*	.034	.24*	.007	−.02	.843
Observability	−.10	.100	−.16	.082	−.03	.704
Psychological modality						
Behavior (vs. Affect)	.01	.216	.00	.736	−.00	.932
Cognition (vs. Affect)	.00	.891	−.00	.824	.00	.933
Desire (vs. Affect)	.00	.795	−.00	.881	.00	.723
Timeframe						
Past (vs. Present)	.01	.215	.04	.057	−.00	.270
Future (vs. Present)	−.00	.454	−.00	.927	.00	.518
Baserate	−.05	.355	−.06	.528	−.05	.533
Baserate extremity	−.09	.108	−.10	.251	−.14	.060
Interpersonal importance	−.03	.558	.10	.277	−.06	.420
Value-relatedness	.00	.946	.15	.092	−.07	.386

Note. Psychological modality and timeframe were dummy coded such that regression weights reflect unstandardized contrasts to the respective reference category. Regression weights for the remaining (continuous) predictors reflect standardized effects. *: $p < .05$.

significant, positive predictor of prediction accuracy in the subset of Likert-scale items but not in the subset of binary forced-choice items. Other than in the main analyses, a negative effect of baserate extremity emerged not only in the multiple regression model but also zero-order. Details on this robustness check are presented in S6 and S7.

Finally, we explored 48 possible (first-order) interaction effects among the considered predictors. Whereas five of these effects were significant at an alpha level of .05 (see S8) and the simple slope plots of these effects suggested that especially effects of item desirability might be moderated by timeframe and modality (see S9), none of these effects survived a more conservative alpha level of .001, which we consider appropriate to guard against Type I errors given the large number of statistical tests. As such, our data contain no convincing

indication of potential interaction effects between item characteristics. However, studies with an even a larger number of items are needed to gain more certainty about this conclusion.

Discussion

It has been a recurring finding in the person perception literature that personality judgments of targets align less with these targets' self-reports when made on evaluative rather than neutral items (De Vries et al., 2016; John & Robins, 1993; Kenny & West, 2010; Wessels et al., 2025). Using a dataset encompassing over 340 thousand first impression judgments made by users of the Who Knows app (Rau et al., 2025) on 296 heterogeneous and mostly

Table 5. Reliabilities and zero-order associations of item features in the complete item pool ($n = 296$)

Feature (Reliability)	1	2	3	4	5	6	7	8	9	10
1: Abstractness (.76)	1	-.03	-.01	.12*	.37*	.13	.14*	-.07	.29*	.15*
2: Desirability (.97)	-.03	1	.20*	.26*	.19*	.22*	.07	-.10	.06	.41*
3: Evaluativeness (.93) [#]	-.01	.21*	1	.06	.18*	.10	.03	-.12*	.60*	.67*
4: Observability (.90)	.15*	.28*	.07	1	.31*	.51*	.39*	.11	.25*	.15*
5: Psychological modality ^{cat}	.42*	.20*	.19*	.33*	1	.39*	.29*	.17*	.19*	.12
6: Timeframe ^{cat}	.15	.23*	.10	.54*	.39*	1	.29*	.06	.05	.08
7: Baserate (.92)	.16*	.07	.04	.43*	.30*	.30*	1	.44*	.24*	.12*
8: Baserate extremity (.75)	-.09	-.11	-.14*	.13	.20*	.07	.53*	1	-.07	-.16*
9: Interpersonal importance (.92) [#]	.35*	.06	.65*	.28*	.20*	.06	.26*	-.08	1	.66*
10: Value-relatedness (.87) [#]	.18*	.45*	.74*	.17*	.13	.08	.13*	-.20*	.74*	1

Note. Reliabilities reflect ICC(3, k) values. Most associations reflect Pearson correlation coefficients. Exceptions are associations involving categorical variables (marked by an uppercase “cat”). These associations reflect η values derived from an analysis of variance (in cases where one of the involved variables is categorical) or Cramer’s V values derived from a χ^2 -Test (in cases where both involved variables are categorical). See S2 for variants of this table based on separate subsamples for Likert-scale and binary force-choice items. Correlations among variables marked with [#] are highlighted by bold print and were deemed large enough to justify combining the respective variables in a composite score. Gray numbers below the diagonal reflect disattenuated associations where reliability for categorical variables was assumed to equal 1. *: $p < .05$.

concrete items, however, we could not find any evidence for evaluativeness undermining the association between self- and other-judgments. In fact, evaluativeness even promoted prediction accuracy in items that were judged on a rating scale from 1 to 10, whereas it was unrelated to prediction accuracy in binary items responded to with Yes or No. However, some caution is warranted when interpreting this finding, as the two item sets also differed slightly with respect to item content, not just with respect to their response format. Besides this, given the rich database, we believe the present results to be robust for novel judges and targets and, within our item pool, generalize across a wide range of concrete, narrowly phrased items. At the same time, the results may not readily generalize to abstract, broad personality items based on adjectives, which are commonly included in short forms of personality inventories and generally

studied more intensely in personality science. Further, our results suggested that the self-report prediction instruction used in Who Knows might have had a more profound impact on how evaluativeness contributed to prediction accuracy than we expected. We next elaborate on the insights into these boundary conditions more deeply and derive possible avenues for future research for each of them.

A first descriptive takeaway is that the overall level of self-report prediction accuracy was modest: Across items, prediction accuracy ranged from near zero to about .25 (on the common r -like metric), and all values were positive, suggesting no item showed systematic mismatches between targets’ self-reports and judges’ predictions. Importantly, our index reflects *relative* accuracy (tracking who differs from whom), not mere base-rate matching. The “small-to-moderate at best” range is theoretically

Table 6. Item features’ unique influences on prediction accuracy by response format

Predictor	All items ($n = 296$)		Scale items ($n = 123$)		Binary items ($n = 173$)	
	b	p	b	p	b	p
Abstractness	-.02	.755	.03	.756	-.03	.693
Desirability	.05	.437	-.03	.784	.13	.122
Composite [#]	.04	.528	.27*	.012	-.09	.251
Observability	-.12	.103	-.28*	.023	-.09	.393
Psychological modality						
Behavior (vs. Affect)	.00	.572	-.01	.368	-.00	.630
Cognition (vs. Affect)	-.00	.608	-.02	.242	-.00	.595
Desire (vs. Affect)	.00	.926	-.02	.437	-.01	.547
Timeframe						
Past (vs. Present)	-.00	.998	-.02	.334	.00	.446
Future (vs. Present)	-.01	.172	-.03	.125	.01	.255
Baserate	.01	.918	-.03	.791	.10	.316
Baserate extremity	-.08	.227	-.06	.543	-.19*	.036

Note. Psychological modality and timeframe were dummy coded such that regression weights reflect unstandardized contrasts to the respective reference category. Regression weights for the remaining (continuous) predictors reflect standardized effects. #: even-weighted composite of evaluativeness, interpersonal importance, and value-relatedness; *: $p < .05$.

informative because it implies that predicting narrow, self-reported personality traces from a brief self-introduction video is a difficult task—even in a very large dataset where item-level estimates are stable. The key puzzle, then, is not whether prediction accuracy exists at all, but why the classic *evaluateness-harms-agreement* regularity did not emerge in this task, and why evaluateness even related positively to prediction accuracy for rating-scale items.

Concrete Items Are Quite Different From Abstract Ones—In Many Regards

A central boundary condition is the item ecology of Who Knows. The item pool was intentionally low in abstractness, and empirically this was extreme: only 9 of the 296 items (<4%) exceeded the midpoint of the abstractness scale. In the same pool, social desirability ratings were roughly unimodal around neutral rather than showing the more bimodal pattern often reported for typical personality-item sets (Anderson, 1968; Britz et al., 2019; Dumas et al., 2002; Leising et al., 2012; Leistner et al., 2025), and evaluateness (absolute desirability) was therefore left-skewed with many items closer to neutrality than to maximal evaluateness. This distributional package matters because it implies that “evaluative” content in this study is being sampled from a different region of the language/trait space than in adjective-based inventories: it is predominantly narrow, concrete, and embedded in specific behaviors and traces rather than broad summaries. As a consequence, item-feature effects estimated in this pool cannot be assumed to generalize directly to classic adjective pools without considering what “evaluateness” and “abstractness” mean in each ecology.

To make the abstractness claim more precise, it helps to distinguish our operational definition from a plausible theoretical interpretation. Here, abstractness was operationalized as perceived interpretive latitude and a major contributor to interpretive latitude is “category breadth” (Hampson et al., 1987): broad descriptors can be instantiated in many different ways, whereas narrow descriptors constrain plausible instantiations. For an abstract descriptor such as “fair,” respondents may think of different concrete referents, such as sharing resources, refraining from exploiting loopholes, avoiding misusing disabled parking spaces, or donating to charity, while still believing that they answered “the same” question. In contrast, more concrete items constrain the range of plausible referents more strongly and may therefore leave less room for such divergence in interpretation.

This clarification also helps to understand the complex relation between abstractness and evaluateness. Within the substance-and-evaluation framework, evaluateness is about the extent to which using a given attribute to judge a target expresses an attitude component (i.e., how strongly judgments on that attribute load on *A*, over and above substance *S*). At the same time, the model explicitly allows attitudes themselves to be partly substance-rooted (e.g.,

targets who actually behave in socially valued ways may elicit more positive attitudes) and partly driven by non-item-specific sources, including generalized positivity or negativity in self- and other-views (Anusic et al., 2009; Furr & Funder, 1998; Heynicke et al., 2022; Rau et al., 2021, 2022; Srivastava et al., 2010; Wood et al., 2010). This is the key place where item concreteness plausibly matters: not because concrete items are “non-evaluative,” but because concreteness can shift the composition of the attitude variance that enters judgments. For narrow statements, any evaluation that enters the judgment is more plausibly tied to the inferred substance implications of that specific act or trace (i.e., “if they do X, that says something socially good/bad”), whereas for abstract trait adjectives, evaluative meaning may additionally arise from the descriptor’s summary-level tone, which can be applied in ways that are partly decoupled from concrete behavioral referents. Put differently, the same absolute desirability/evaluateness score may reflect different mixtures of (i) substance-mediated evaluation versus (ii) descriptor-level evaluative tone, depending on where the item sits on the abstract–concrete continuum.

The multifaceted nature of abstractness may also help explain why it was comparatively hard to rate reliably (with inter-rater agreement improving only after excluding a subset of raters who likely misunderstood the instruction), a difficulty noted elsewhere as well (Leistner et al., 2025). A plausible reason is that “room for interpretation” bundles multiple components—most notably category breadth and semantic vagueness/connotation—so raters may weight different aspects. Future studies could reduce this ambiguity by explicitly steering raters toward category breadth.

Finally, this framework resembles the “bandwidth–fidelity tradeoff” (Cronbach & Gleser, 1957; Ones & Viswesvaran, 1996). Abstract items have higher bandwidth but lower semantic precision in what is being referenced, whereas concrete items have lower bandwidth but constrain meaning more strongly by pointing to a specific act or trace. In thin-slice video contexts, these forces can work against each other: abstract items may be more likely to afford at least some weakly informative cue, whereas concrete items can yield sharply informative cues when the relevant information is present in the clip but often yield little usable information when it is not. Consistent with this theoretical ambiguity (and the strong restriction of range in our item pool), abstractness was essentially unrelated to self-report prediction accuracy in the present data. Taken together, these points sharpen the “different region of language space” argument in a way that is directly relevant to the evaluateness puzzle and supports treating our findings as boundary-condition evidence rather than a straightforward contradiction of the earlier literature.

Judges Might Be Able to Detect Which Targets Self-Protect More Than Others

A second boundary condition concerns task framing. In Who Knows, judges were instructed to predict targets’

self-reports rather than to provide their own impressions. Agreement with self-reports can therefore arise not only from tracking targets' substantive differences, but also from anticipating how targets describe themselves on evaluative content. In our formalization (Figure 3), this is represented by an additional pathway: judges' predictions may partly align with stable between-target differences in self-protection tendencies, which should matter more when items are evaluative and self-protection incentives are higher. This mechanism does not require strong inference about private motives; it only requires that stable differences in how favorably targets respond can be inferred to some degree from information available in a short self-introduction.

Our exploratory favorability-alignment analysis provides evidence consistent with this possibility. Targets differed systematically in the favorability of their self-reports across items ($ICC = .150/.078$), and targets also differed in the favorability of the self-reports judges predicted for them ($ICC = .040/.017$). The target-level components correlated positively ($r = .49/.31$), indicating that targets who described themselves more favorably also tended to elicit more favorable predicted self-reports. This pattern is compatible with the interpretation that judges exploit between-target differences in the tendency to respond self-protectively when forming self-report predictions. Such tendencies could contribute to prediction accuracy especially for evaluative items: If a target generally tends to describe themselves favorably, judges should predict stronger endorsement of desirable content and weaker endorsement of undesirable content. For neutral items, by contrast, the same target-level tendency provides little guidance because the response options do not clearly differ in favorability.

Alternatively, the alignment might arise because some targets possess more substantive characteristics that are commonly evaluated positively by perceivers ($S \rightarrow A$), and those same targets also provide more favorable self-reports. These routes are not mutually exclusive: Judges may rely on item-relevant cues, broader target impressions, cues of self-protection, or generalized positivity toward the target, with their relative influence depending on which cues are actually available in the video. Notably, the self-report prediction instruction makes the self-protection route especially plausible in the present paradigm: Judges were not merely asked what the target is like, but how the target would describe themselves. Moreover, because the task was video-based, judges could draw on visible cues to targets' self-protective style, including appearance, clothing, nonverbal behavior, and general presentation on camera. This may help explain why effects of item evaluativeness differed from prior self-other agreement findings. However, to cleanly disentangle these routes, future research requires designs that (a) measure targets' self-protection tendencies independently (e.g., socially desirable responding, impression management scales, or observer-coded self-presentational behavior) and (b) separate judges' predicted self-reports from their evaluative impressions within the same task.

Why All This Emerged for Likert-Type Items but Not for Binary Items

Evaluativeness was associated with prediction accuracy for Likert-type items but not for binary items. A possible explanation for this concerns graded evaluative self-reporting. Rating scales allow small shifts toward the desirable pole while remaining plausible, whereas binary items require categorical commitments and may therefore reduce both the prevalence and variability of evaluative self-protection. Under this account, evaluative self-protection should create stronger stable between-target variance for Likert items, which judges can exploit when predicting self-reports. Descriptively, the favorability-alignment results are consistent with this possibility (cf. Table 2; larger target ICCs and stronger target-level self-other favorability correlations for Likert than binary), although these results do not identify the mechanism.

A second explanation has to do with response base rates. For many binary items, responses were highly skewed (e.g., 90% "No"). In such cases, the item contains little person-to-person variation, so there is limited room for any judge to track individual differences, and agreement is constrained. In our results, this constraint shows up clearly: In the binary models, base-rate extremity was the only unique predictor—that is, the only predictor that was associated with accuracy after adjusting for the remaining item features—and it predicted *lower* accuracy ($b = -.19, p = .036$); in the Likert models, by contrast, the evaluativeness/importance/value composite predicted accuracy ($b = .27, p = .012$), and base-rate extremity did not add explanatory power.

A third explanation is methodological: binary and continuous outcomes were modeled with different link functions and then mapped onto a common r -like metric, which can attenuate detectable moderation in the binary case. These explanations are not mutually exclusive. A direct test would manipulate response format for matched content, thereby separating response granularity from item content.

Why Did Most Other Item Features Not Matter?

Beyond evaluativeness (in Likert-type items) and base-rate extremity (in binary items), most other item features were weak or inconsistent predictors of self-report prediction accuracy, and none of the explored two-way interactions survived correction for multiple testing. This suggests that, in a thin-slice self-report prediction task with narrow concrete items, commonly studied item features (e.g., modality, timeframe, abstractness) do not translate into robust item-level moderators of prediction accuracy in this item population. One coefficient warrants cautious attention: observability showed a negative unique association with prediction accuracy in the Likert multiple regression despite weak zero-order associations. This pattern is consistent with a suppression effect driven by the strong overlap of observability with timeframe and psychological modality. Once differences between items

pertaining to different timeframes (past vs present vs future) and items pertaining to different modality categories (affect vs behavior vs cognition vs desire) are held constant, the remaining variance in observability may capture item content that is *rated* as observable in principle but is not well represented in a brief self-introduction video. Clarifying the role of observability requires stimulus sets with controlled cue availability (e.g., behavior-eliciting videos or longer interactions), rather than relying on generic self-introductions.

Limitations and Future Directions

Several limitations need to be acknowledged. First, the item pool analyzed in this work is not intended to represent personality items broadly. Instead, it represents a population of narrow, phrased items suitable for a gamified first-impression task, and the unimodal desirability distribution observed here is likely characteristic of that population. Second, the items' response formats were chosen to match item content, so response format is partially confounded with content. Third, between-target analyses remain constrained by the number of targets, which limits strong claims about which target characteristics systematically increase or decrease predictability. Fourth, the feedback mechanism of the Who Knows app may have allowed users to learn recurring aspects of the task over time. Because users received feedback after each round, highly active users could gradually acquire knowledge about common response patterns, extreme item base rates, or recurring targets. Target-item combinations never repeated, so users could not simply memorize correct responses for specific target-item pairs. Still, repeated exposure may have increased reliance on learned response norms or target impressions.

A final central limitation is that the two boundary conditions emphasized here—high item concreteness and self-report prediction instructions—were fully confounded in the present design. Thus, the present data cannot determine whether the observed effects of evaluativeness are due primarily to item concreteness, the self-report prediction instruction, or their combination. Future work could disentangle these factors using a paradigm in which cue availability is deliberately engineered rather than incidental. Within such a stimulus set, three design factors can be crossed: (a) item abstractness (broad adjectives vs narrow concrete statements), (b) instruction (impression ratings vs self-report prediction), and (c) response format (graded vs binary). Stimuli such as longer interaction segments or behavior-eliciting recordings would allow testing whether evaluativeness effects depend on the availability of item-relevant cues and on the incentives created by prediction instructions. Adding independent measures of self-protective tendencies would further adjudicate whether evaluativeness amplifies a self-protection pathway, a general positivity pathway, or both.

A related but conceptually distinct direction concerns interjudge consensus (other–other agreement). In Who Knows-type paradigms with first impressions from brief

videos and highly concrete items, it would be informative to examine which item properties elicit stronger consensus among judges, and whether such consensus reflects shared diagnostic cues versus shared stereotypes or response norms. Importantly, high consensus need not imply high correspondence to targets' self-reports; thus, consensus should be treated as a separate outcome with its own substantive interpretation.

Another complementary direction is to use paradigms that randomly cross judges, targets, and items (as in Who Knows) to move beyond item-level moderation and examine stable individual differences in judges' ability and targets' "readability," as well as their interactions with item properties. Such judge $\times$ item and target $\times$ item interactions map onto Funder's (1995) concepts of *expertise* and *palpability*, respectively, and would help identify which items are especially diagnostic for high-ability judges or especially revealing for highly readable targets.

Conclusion

In sum, in a large thin-slice dataset of self-report predictions about narrow, concrete items, evaluativeness did not undermine the judges ability to align with targets' self-reports and was even positively associated with prediction accuracy for graded response formats. The broader implication is that item-feature effects are properties of an item population embedded in a specific judgment task. Theories and models of person perception that seek to predict when agreement will be high or low should therefore incorporate both the nature of the item pool and the goal induced by the instruction, particularly when moving from adjective inventories to concrete behavioral statements and from impression ratings to self-report prediction tasks.

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ORCID iDs

Richard Rau  <https://orcid.org/0000-0002-1490-8584>

Sebastian Kueppers  <https://orcid.org/0009-0004-6702-5154>

Michael P. Grosz  <https://orcid.org/0000-0002-1949-4384>

Ethical Considerations

All procedures described in the present research were approved by the ethics committee of the University of Münster (protocol number 2021-25-RR).

Consent to Participate

All subjects involved in this research provided informed consent prior to participation.

Consent for Publication

All identifiable subjects provided consent for publication.

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Data Availability Statement

All data and analysis code to reproduce results are available under <https://osf.io/rjs2u/>.

Open Science Statement



The study materials, data and analysis scripts used for this article can be accessed at <https://osf.io/rjs2u/>.

Notes

1. This zero-order coefficient is not reported in the published article but was personally communicated to the first author by David A. Kenny.
2. The SOKA model also makes a complementary case for the item feature of “observability” which is assumed to shape self–other agreement by affecting the substance saturation of the *other*-report while the self-report is assumed to be largely unaffected given that self-reports hinge mostly on stored self-knowledge rather than direct self-observation. We come back to the item feature of observability at a later point of this manuscript.
3. As can be seen in Figure 1, the Who Knows app also uses two further types of items which are not considered here. These include trivia questions (e.g., “Who or what is ‘Banksy’?”) and open-ended questions (e.g., “Which of your traits gets on your friends’ nerves?”). Trivia questions were not considered here because individual differences in knowledge are not encompassed in most conceptions of personality. Open-ended questions were not considered because the Who Knows app uses target-specific distractors for these items (e.g., the response options for the annoying trait item could be “cynical,” “talking too much,” “always late,” and “domineering” for one target and “pessimistic,” “cynical,” “silent,” and “meticulous” for another target). As such, self–other agreement for these items varies between target*item pairs rather than between items.
4. These raters showed both a lack of agreement, $ICC(3,10) = .26$, and a markedly high mean rating ($M = 2.3$ as compared to $M = 1.3$ found for the remaining raters). The high mean rating suggests that even though the item-pool had been deliberately generated to include mostly concrete items, these raters apparently made use of the entire scale from 0 (very concrete) to 5 (very abstract), perhaps because using the entire scale is sometimes encouraged in other questionnaires with Likert scales.
5. To account for the complex, cross-classified data structure, the model also included random intercepts for items, judges, targets

and target*item pairs. For an in-depth elaboration of this type of crossed-random effects model, see Kueppers et al. (2024).

6. Without excluding raters who hardly agreed with one another concerning the items’ abstractness, the mean abstractness value would have been slightly greater ($M = 1.64$) but still, very few items would have fallen above 2.5 (11 instead of 9 items, still $< 4\%$).

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Appendix

Appendix A.

Instructions Used in Previous Research to Assess the Observability of Items

Reference	Instructions for raters rating observability	Critique
John and Robins (1993)	“Some traits refer to behaviors that can be easily observed by an outside observer. Other traits refer to behaviors that can be observed only by person himself or herself.” Raters then provided ratings on a scale from “extremely difficult to observe by an outside observer” to “extremely difficult to observe by an outside observer.”	Traits are defined exclusively in terms of behaviors. The possibility of traits (or items) pertaining to affect, cognition, or desires, is not considered.
Kenny and West (2010)	Raters were provided with a description of the context in which participants in the original studies had provided personality judgments of targets. They were then instructed as follows: “Do others have sufficient information to be able to make a valid judgment of the target?”. They then provided ratings on a scale from “not at all” to “completely.”	Observability is defined not as a feature of the item but in terms of the information available to judges in the specific study context.
De Vries et al. (2016)	Some aspects of personality are easy to judge by external observers, whereas some aspects may be judgeable only by people themselves. For each item, please indicate how easy it would be for an external observer to decide if it describes the person being rated.’ Raters then provided ratings on a scale from “very difficult” to “very easy.”	Observability is defined not as a feature of the item but in terms of a state of the judge (ease of judgment).
Wessels et al. (2025)	“How well can the respective characteristic be observed from the outside?” Raters then provided ratings on a scale from “not observable at all” to “very well observable.”	None